

Aayush Kumar Dubey

Bachelor of Technology (B.Tech)
Instrumentation and Control Engineering
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GitHub Profile

LinkedIn Profile

EDUCATION

- Netaji Subhas University of Technology, New Delhi (Formerly NSIT)** 2023-2027
Bachelor of Technology (4 Year) in Instrumentation and Control Engineering

EXPERIENCE / CERTIFICATION

- IIT Guwahati (Online Data Science Program)** Jan 2025 - April 2026
Trainee, FusionX-3
 - Designed and implemented a triple-stream CNN (RGB, DCT, PRNU) with a patch-wise Xception backbone (1×1 conv heads) to capture localized forensic artifacts for scalable AI-generated image detection.
 - Evaluated on 13k Artifact Mini samples, achieving 65% accuracy and 0.61 F1-score, outperforming reported baseline detectors on this dataset and demonstrating robustness to unseen artifact distributions and post-processing noise.
- Tata AI Collections Strategy Simulation | [Certificate](#)** May 2026
Job Simulation / Forage
 - Performed EDA on financial delinquency datasets, handled missing values/outliers, identified key risk indicators, and analyzed customer repayment behavior using predictive analytics concepts.
 - Developed a Logistic Regression-based delinquency prediction strategy with Accuracy, Precision, Recall, F1, and AUC-ROC evaluation planning, while proposing AI-driven collections workflows aimed at reducing delinquency rates, improving repayment efficiency, and enabling scalable decision-making.

PROJECTS

- KAI — Knowledge Augmented Intelligence | [Link](#)** Apr 2026
GenAI, LangChain, FastAPI, React
 - Built an AI-powered Retrieval-Augmented Generation (RAG) chatbot capable of answering queries from uploaded PDF documents using semantic retrieval and Google Gemini.
 - Designed a modular FastAPI backend handling document ingestion, chunking, embedding generation, and vector-based context retrieval workflows.
 - Developed an interactive React frontend enabling real-time conversational querying and seamless communication with AI backend services.
 - Implemented context-aware retrieval pipelines using LangChain to improve relevance and grounding of LLM-generated responses.
- FusionX-3: AI-Generated Image Detection** Sep 2025
Triple-stream deep learning model for AI-generated image detection with cross-dataset evaluation
 - Developed a production-style multi-input CNN system using RGB, DCT, and PRNU forensic features, training on 120,000 images and achieving 95% accuracy and 0.95 F1-score, matching state-of-the-art deepfake detection benchmarks.
 - Implemented a patch-wise Xception-based architecture (RF=43) with 1×1 convolutional classifiers, enabling localized artifact detection and reducing overfitting, with validation loss minimized to 0.12 using Adam optimization.

TECHNICAL SKILLS AND INTERESTS

Languages: C++, Python, HTML, CSS, SQL

ML & AI: GenAI, RAG, LangChain, TensorFlow, Scikit-learn, OpenCV, NLP, LLMs

Frameworks: FastAPI, REST APIs

Developer Tools: Git, GitHub, VS Code

CS Fundamentals: Data Structures & Algorithms, OOPs, DBMS

POSITION OF RESPONSIBILITY

- Logistics Coordinator — MOKSHA 2025 - The Annual Cultural Fest of NSUT** March 2025
 - Coordinated logistics for 300+ volunteers and managed sponsor activations during NSUT's annual cultural festival.
 - Successfully executed all sponsor deliverables for Moksha 2025, reinforcing brand partnerships and event visibility.

ACHIEVEMENTS

- Ranked in the top 4.5% among ~1 million candidates in JEE Main 2023.
- Orchestrated educational workshops for 75+ underprivileged children, focusing on literacy and numeracy skills, resulting in a 40% improvement in test scores within three months.